

Learning Rate (4):

Recommended LR for Cross-entropy loss is in the range 0.01 to 0.1.

Our model performs best with higher LR, around 1.0.

Consider the cost of a single output neuron for Cross-entropy loss vs. MSE

• Note that $\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z)) = a(1-a)$

$$C_{\text{MSE}} = (a - y)^2 \rightarrow \frac{\partial C}{\partial a} = 2(a - y)$$

$$\rightarrow \frac{\partial C}{\partial z} = 2(a - y) \cdot \sigma'(z) = 2(a - y) \cdot a(1-a)$$

$$C_{\text{CE}} = -[y \ln(a) + (1-y) \ln(1-a)]$$

$$\rightarrow \frac{\partial C}{\partial a} = -\frac{y}{a} + \frac{1-y}{1-a} = \frac{a(1-y)}{a(1-a)} - \frac{y(1-a)}{a(1-a)}$$

$$= \frac{a - ay - y + ay}{a(1-a)} = \frac{a - y}{a(1-a)}$$

$$\rightarrow \frac{\partial C}{\partial z} = \frac{a - y}{a(1-a)} \cdot \sigma'(z) = \frac{a - y}{a(1-a)} \cdot a(1-a) = a - y$$